active business consulting agency

AID-mh_2.7.1.7

Focus: 2G, 3G, 4G, GSM, IMS, UMTS, SIP, VoIP, NGN, LTE, Network, Juniper

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Date: 06. February 2014

Personal Data

Name:	AID-mh_2.7.1.7
Year of Birth:	1956
Nationality:	German
Languages:	German, English, Urdu/Hindi
IT Experience since:	1984
Available in:	Worldwide
Education:	Kingston University, England Sept. 79 to July 82: B.Sc. (Hon) Applied Science with Computing. Oxford University, England Sept. 96 to Sept. 98: M.Sc. (part time) in Software Engineering. Software Formal Specifications, Design, Critical Systems engineering, Advance Software development, Concurrency and Distribute Systems, Software development Management, and Software Engineering Mathematics. OOD Design Patterns using Java.

Competence

Responsibilities for 3G UMTS IMS LTE VoIP Network Management Architecture Specifications, Design, and Development.
Member of British Computer Society (MBCS), IEEE, and IEEE Communication Society.

Branches

Telecommunication - for 25 Years: Alcatel, Nokia, Siemens, Ericsson and Intel.

Hardware

Ascii/X - Terminals
embedded Systeme
Rational
Siemens MX
SUN
Texas Instruments
UNISYS

Programming

ABAP4
ASN.1
Assembler
Basic
C
C++
Clipper
CLIPS
Cobol
CodeWarrior
Coral
dBase
DCL
Ebus-Generator-C
Eiffel
Emacs
ESQL/C
extra
FOCUS
Forth
Imake, GNU-Make, Make-Maker etc...
Java
JavaScript
MATLAB / Simulink
Paradox
Pascal
Perl
PHP
PL/SQL
PL/1
Powerbuilder
Psion OPL
Python
Qt
SDL
Shell
Tcl/Tk
TTCN

4gl

Operating Systems

MS-DOS
Novell
OSF/Motif
OS/2
PalmOS
RTOS (Real Time OS)
Siemens ORG-R/M (BS300)
SUN OS, Solaris
Unix
VM
VxWorks
Windows
Windows CE

Databases

Access
IMS
JDBC
Object Store
xBase

Network

ATM
BSC
CORBA
DCAM/ISAM
FDDI
FTAM
HDLC
HDSL
IMS/DC
Internet, Intranet
ISDN
ISO/OSI
LAN, LAN Manager
MMS
OSF/DCE
OS/2 Netzwerk
Public Networks
RFC
Router
RPC

SNMP
TCP/IP
TIP
Voice
X.400 X.25 X.225 X.75...

Products | Standards | Experiences

Juniper Firewalls SRX240 , EX4xxx, EX32xx, MX, Junos
Data Center Network Design, Firewalls, Riverbed, LAN/WA
JNCIP/JNCIE/JNCISLevel, VMWare ESX Server
Cisco Secure Manager, Netflow,Cisco virtul Nexus ASA, ACE
Juniper JNCIP/JNCIE/JNCISLevel
Cisco Secure Manager, Netflow, NetScreen Security Manager
.NET Platform , Jenkins, GIT, Linux, Jira
Cisco Juniper EX MX, SRX, Routers Switches
IPTV, DSLAM, BRAS, ISP, L2TP, PPPoE
BGP, Junipers, OSPF, EIGRP, QoS, VoIP, VPN SSL
Network Management LTE,GSM, UMTS Protocols, CS/PS Networks
VoIP, LTE, 4G, IMS, RTP, SIP, SDP, RTP, CS, PS Core Networks
HSDPA, HSUPA, RLC/MAC L1,L2, L3 RLC, PTT, Presence PDC
UTRAN WCDMA, WiMAX, System Test and Verification
SIP, RTP, SDP, VoIP, MMC,CMS, Node B, BTS, RRC, RNC, L2/3
3G, GSM, GPRS, LTE, 4G, HSPA, GGSN, Radius, IMS, X-CSCF
Cisco Voip PABX, CUME , Cube, VLAN
Cisco Secure Manager, CCNA, CCPA, CCIE, Netflow, NetScreen
Cisco Routers and Switches, & ISG Juniper Firewalls and WAAS
VoIP, IP, SIP, DHCP, DNS, IPV6, NAT, VLAN
3G, GSM, GPRS, LTE, 4G, HSDPA, IMS, CSCF, SBC
RTP, SDP, MMC,CMS, Node B, BTS, RRC, RNC, L2/3
HSDPA, HSUPA, RLC/MAC L2/L3 RLC, PTT, Presence PDCP
UTRAN WCDMA, WiMAX, KPI, System Test and Verification
Cisco Call Manager V7.x, V8.x, CUBE, Routers, Gateways Firewalls
Avaya CM 5.x, SM, Alcatel F700 R7.x , F750 R9.x
MMTel, CMS, HSPA, AMR, RTP, SDP, SIP, PTT, Presence
MPLS/GMPLS IP Routing prototcols BGP-4, OSPF, IS-IS , RIP
TCP/IP IPv4, IPv6 , VPN, DiffServ, IPSec
SDH, ATM, IP ,ISDN PRI, PSTN,DSL, xDSL, DSP
OS Symbian C++ , Java J2ME MIDP, MMS , S60/80/90, MTM, UIQ
Java J2ME MIDP, CLDC, MTM, UIQ Messaging
LTE, 4G, IMS, RTP, SIP, SDP, RTP,, CS, PS Core Networks
MetroWerks CodeWarrior, MS Visual C++, Continus, CM Synergy
IPTV, DSLAM, BRAS, ISP, L2TP, PPPoE
Network Management Specifications, Software Architecture,Design
Network Management TMN FCAPS SW Development, Unix, C/C++, Java
Network Management Architecture, OSI Stacks, SNMP, SOAP
Sun Solaris, Unix, HP UX, C, C++, Shell Programming
J2EE Client/Server, Java Web Server, J2ME(MIDP, Java) PDA, Linux
BEA WebLogic, Java Script, JSP, EJB, JDBC, XML, HTML , HTTP
Apache Tomcat Web Server, Java Servlets, Java Server Pages
Object Oriented (OO) Analysis and Design, C++, ASN.1, GDMO
Telecommunications, PDH, SDH, ITU, ISO, TMN , POTS, ISDN, V5.
Software Process Improvement (SPI), QA, ISO9001, IEEE, CMM .
TTCN/Telelogic, Rational RequisitePro, LabView,,TT Workbench
Rumbaugh, Shlaer-Meler, Booch Rational Rose/C++
Project Management (various projects), Microsoft Project
HP-UX , XMP/XOM, CMOT, CMISE , UML, Rational Rose, SDL/SDT

Structured Analysis/Design, Formal Methods: Z, Yourdon
OSF Motif, Windows, HP OpenView 6.x , NT 4.0
Oracle Database (RDBM), V7.0, SQL*Plus, MySQL, PL/SQL, ProC
Test tools, Test bed, Generic Test Harness, PowerPilot, TestNet
System Testing: PreIntegration, Module, Qualifications , Acceptance
Test Simulation/ Traffic Generation Tools for GSM/GPRS/UMTS
Rational Suite: RUP, Rational Rose(UML), ClearCase, ClearQuest
GNU Tool chain,CME, Eclipse IDE, Lauterbach, EtherReal(Wireshark)
Linux, OmniTracker, Catapult, and Pegasus , DOORS, SOAP
NetAct OMS, FlexiBTS, IP Router
IN, JAIN, NGIN/IMS, X-CSCF AS
GNU Tool chain, JIRA, PERFORCE, Eclipse IDE, Lauterbach JTAG32
Cygwin, Moses, SDE, SDK, SDP, SIP, RTP, CME, Windows Mobile OS
SAP J2EE NBAP Portal Enterprise NetWeaver Environment
Agilent Signaling analyzer, RhodeSchwarz
Cisco Voip PABX, CUME , CheckPoint & ISG Juniper Firewalls, VLAN
Cisco Secure Manager, NetScreen Security Manager
BGP, OSPF, QoS, VoIP, VPN (SSL, IPSec), Firewalling
Project Management, ISTQB, CCMI, Prince II
Agile, SCRUM, ITIL
LLDM, LTE MA Trace Analysis
IRAT IOT ,Artemis,Mobile Analysier(L1U, URRC)
JIRA, Embedded Linux, Andoird, C++
Virtualization VMWare vCenter, VCloud, VMs
Data Centre, BGPEX/MX Juniper CiscoRouter Switches IPv4/IPV6
Juniper Firewalls SRX240 , EX4xxx, EX32xx, MX, Junos
Data Center Network Design, Firewalls, Riverbed, LAN/WAN
JNCIP/JNCIE/JNCISLevel, VMWare ESX Server
Cisco Secure Manager, Netflow,Cisco virtul Nexus, Virtual ASA
Cisco/Juniper Routersand Switches, LDAP/AD, ISS
BGP, OSPF, EIGRP, QoS, VoIP, VPN (SSL, IPSec)
Oracle Database(RDBM), MySQL, MS SQL Server 2008, AD, DNS

Projects

Year: 08/2010 – dato
Customer: Intel Mobile Comunication

Project Description: I am a System Engineer responsible for system analysis, feature development, debugging, integration, test execution , and coordination of RF RAN Modules, IRAT IOT verification of all official customer releases. I perform the verification/Conformance acceptance testing in Lab/Live for VoIP, 2G, 3G UMTS, Node-B and LTE(telephony) Networks. Afterwards, I analyse debug all issues arising from AP platfrom based on Android, Modem based on Linux real time kernel driver , using the in house Mobile Analyser test tracing utilities and System Debugger Lauterbach(T32).
Some of the protocols and tools used were MySQL,MS SQLServer 2008, MS AD, DNS , SIP VoIP VoLTE, CSCF-P, SDP, IPSec,Sigcom, RTP and BGP overVoIPnetworks. Performance optimisation and System Stability was done using MYSQL and other in house Network tools
Together with the solutions analysts, I validate that the business requirements and scope have been analyzed and documented. Afterwards, I present requirements, scope and design artifacts to management for approval. The target is based on ARM 11 DSP Architecture for 2G/3G LTE overHSI Mux interfacesbetween the Modem, RIL, and the integrated Andorid API. I followed up and analysed all the issues using MA, Artemis, and LLDM Trace Utility. For Real Time debugging I used Lauterbach T32 Debugger for indeapth Real Time onchip analysis, and Optimization. Other tools used on the project are ClearCase OptiCM 5, Unix (Urania), Citrix, Embedded Linux, Android, C, and C++.
I get involve hands on with implementation and execution of end to end Test Case scenarios. This involves using latest test tools and utilities flashing of official releases, on theTarget HW. To performSITprior to handing over to the IRAT test teams for more details test analysis. This includes use ofLauterbach Trace 32,MAmessage decoding andLLDMtrace analysis.

Year: 08/2008 – 08/2010
Customer: Belgacom/Proximus

Project Description: I am responsible for 3GPP System Architecture specifications,Planning andOptimisationof 2G, 3G HSPA LTE, SGSN, IMS, GSM, UMTS, WCDMA, WIMAX, RAN VoIP applicationsforTelecom, Security, and Core Networkdomains.

I participate in deeper analysis, and development of DSP RTOS , LTE software protocols, including testing of LTE mobile RAN/Core Protocol Stacks(L1/L2/L3) for Converged Services.
The TCP-IP VoIP Session Border Controller, Proxy Server, Telephony solution is primarily based on PABX: Avaya, Cisco CUCM , CCME, with CUBE and CCM 7.1 over Cisco Routers and Switches, & ISG Juniper Firewalls PKI , PGP , BGF, OSPF, IS-IS. Participated in Cisco Certification program at CCNA/CCNP levels.
I certified VoIP integrated solutions for IMS Core, NGN Fix, NGN Mobile(Proximus) , Data and Billing applications. I participated CRM process.
For configuration and deployment , I have used the following SW development, test and Integration tools: Matlab, C, C++, Java, CORBA, SOAP, ULM, Linux, IP, SIP, DHCP, DNS, NAT, OOD, UML, GUI Framework (Opensource), Clearcase, CME, CM Synergy Eclipse, DebugMux, Python, Tcl, Perl, Lauterbach , QA Tools and Ethereal (Wireshark) Trace analysis. Agilent Signaling analyzer for Wireless/network traffic, together with Rhode Schwarz.
I Manage all Parallel Testing Tracks and to organize and Track the testing work efficiently according to the Test Plans. My tasks work autonomously to interact with the technical teams and with external vendors. I am responsible for Coordination, communication and reporting of all issues to the stakeholders.
I support the IMS architecute leves to integrate the VoIP IP-PABX solutions to all CSCFs , LSM, BGF with Acme SBC over Fixed Access, and MGX .
For the proposed solutions, I am required to Configure the lab test environment to connect the IP-PABX via SIP trunk to the network VoIP platform and proceed to the testing of the IP-PABX SIP trunking following a predefined test list and take the related call flows traces.
All tests results are docummented, I a nalyze and diagnose the issues that appeared during the test period . To Validate the analysis with the concerned vendors.
I proceed to “arbitrage” of the issues by determining their owner (IP-PABX or Network VoIP platform vendor) in order to come to a resolution of the problems.
Afterwrds, I re-test the call flows concerned by the issues when patches/upgrades are provided by the vendors. After succesfull test results, I Support Operational department when SIP trunking issues are to be investigated in depth.
I Create the trouble tickets to the vendors and assure the follow-up and validation of the fixes.

Year: 10/2006 – dato
Customer: Sony Ericsson

Project Description: I am responsible for defining RAN Mobile 3G IMS UMTS, VoIP SW Application and Platform Architecture.
My tasks include defining System Requirement specifications, Architecture and software Solution for implementations of the Non Access Stratum on Ericsson Next Generation Networks(NGN) for 3G and 4G(LTE). I developed SW modules within Mobility Management, Call/Connection Management and Session Management.
I supported design of overall end-to-end architecture solution and planning, Development, Porting and verifications of complete product. I defined 3rd party vendor technical acceptance model as part of the System activities.
The embedded SW modules which I was responsible provided solution for 3G, GSM/GPRS, IMS, UMTS, HSPA, OSS, VoIP call, Video, Instant Messaging, POC/PTT, Presence, with Converged Multimedia Services on a Mobile Terminal Platform.
The Presence Services were developed using standard OMA Schemas to support full Group and List Management across the client server networks, to support end-to-end presence for all user . This solution was presented in Barcelona Mobile exhibition in Feb 2008.
System Test activities included supporting debugging and troubleshooting of SIP, RTP, SDP, RLC MAC, Interface Protocols, and configuration of IN/NGN, IMS, X-CSCF Application Servers Network Platform. To analyze, and to make recommendation to resolving problem in the lab and in the filed.
For development of Embedded RT SW tasks, I was involved with using C, C++, UML , SDL, Clearcase, Matlab, CME, CM Syngery, Eclipse, Lauterbach , Fido, Win CE, and Ethereal for analysis of IMS traces.
Nokia- Siemens Germany August 2005 - October 2006
Defined 3G RNC Signaling, HSDAP, HSUPA, and IPSEC on WiMAX BTS. Analyse KPI parameters for System Dimensioning, Transmission, and Optimization for Startup, System Recovery and Escalation strategies.
Prior to this project, I developed 3G Ethernet, IP Network Router, IPSec security, IMS, and DCN Information Model . I participated in subsystem SW development activities and testing on embedded LTE, ARM, DSP targets.
Design and development of RNC Node B, and BTS SW, for both Mobile WCDMA, and UMTS Networks.
Design of L2&L3 signaling protocol application, together with Multimedia IMS, and VoIP protocols.

In addition, defined Tracing, and Error diagnostic mechanism for the RNC . This was used to support different levels of Testing, and Integration activities, both in lab and in the field.

Year: 01/2002 – 08/2005
Customer: Nokia - Helsinki Finland

Project Description: I have been responsible for SW Specifications, Architect, Development, Test and Integration activities on Mobile 2G GPRS, 3G UMTS/W-CDMA Embedded SW Push to Talk over Celluar Project. The project was based on Symbian Series 60/80/90 PTT. The applications are based on Symbian vOS 6.0/7.0 OS running on various Realtime Embedded Products, including the Application Engine (UI) browser for Mobile Terminal devices.
All applications developed in C++ CodeWarrior, MS Visual C++, Continus, CM Synergy and Clearcase, Eclipse IDE, Lauterbach. All testing including Unit testing done to fine tune for Memeory and performance optimations.
The core technologies and Network used were, 3GPP R5, IMS, 3G Terminal in SIP architecture, RNC WCDMA Control(CS, PS Core networks RRC, RRM), RTP, SDP, MMF, and XML.
The Software was developed as a Server to take care of session management with SIP and SDP-protocols and it controls PoC plug-in of the multimedia framework (MMF), which handles data transfers with RTP. It also provides an API for applications wishing to use PoC Terminal Device.
. The SW was developed using using Rational Rose EPOC C++, UML, OOD, CodeWarrior, and Continus/CM Synergy CCM.
I assisted in defined and implemented a SW Development and a Test Process. Testing of Functional and Performance requirements, was done various Tools and utilities including, QA Load, QA Runner, LoadRunner, WinRunner, Test Director, HttpUnit, and Junit. All testing was performed according to the Internal Test Process.

Year: 05/2000 – 07/2001
Customer: Tellabs

Project Description: I was Senior Software Engineer with responsibility on Network Management System FCAPS Specifications for Requirements, Architecture, design and Implementation of UMTS 3G , Mobile IP , ATM/IP, Mentoring, and Internet Applications.

I specified and designed the components necessary for generic Application Platform (SNMP Proxy Agent/Server, CLI), which support IP Protocols (Signalling, Routing). This project is part of 3G UMTS Core Network Level for Mobile IP, and Fixed Access Networks Applications. This Software is distributed and integrated on Access Node (AN), Network Terminal Unit (NTU), and Integrated Access Node(IAD). The infra structure for these Access devices is IP/ MPLS Core to support VPN.
My SW development responsibility, was for implementing VoIP and ATM SW Switching, Routing, UNI/PNNI for PVCs, and SVC, and VPN. The IP applications supported the requirement for the Internet Access, and the Mobile IP applications. The SW provides support for point-to-point and point-to-multipoint connections. The VoIP for Routing protocols included BGP-4, OSPF-2, IS-IS and RIP, corresponding Asics were developed to support switching. The IP applications supported QoS, based on DiffServ and MPLS.

Year: 08/1996 – 05/2000
Customer: NOKIA

Project Description: I have been working in the System Concepts groups for the Nokia Telecommunications (NTC). My role has been to define the system concepts for the Mobile and Fixed Access Applications, using some of the ITU, and Nokia specific standards. The technical specifications were used to determine the architecture choice(s) for the Access Node with Multiservice applications.
Some of the main tasks included: Information models for the Network Management over the Q3, System Requirements, System Functional Specifications, System Test strategy, Design of Major SW components, and defining the Core Cross Connection subsystems for all type(s) of signals through the Access Node.
My Development role included being responsible for a subsystem and all its classes, and their interfaces. This software was Real Time, Embedded using OO methodologies, , UML, OMT, Rational Rose, C/C++. A more formal use of the GDMO Modelling , MIB and the ASN.1 were used throughout these projects: from specification to implementation. The internal and external communication between these objects was implemented using generic Message Interface classes.
Supported activities for Multimedia Wireless smartphone applications, on OS Symbian EPOC Platform. The applications used most of OS Symbian Kernel Components, including Application Framework, Communication Infrastructure, Network Stacks,

Connectivity Plug-Ins , SyncML, and Application Protocols engines. The applications were developed in Real Time embedded environment, using Object-Oriented architecture, memory management, event handling mechanisms, and , multitasking. Using Open (J2ME, CLDC/MIDP, C++, and PersonalJava) as a native environment.
I was responsible for developing test components based J2EE Multitiered Architecture on both Client and Servers . The implementation used XML, Java JSP Servelts, JSP, BEA WebLogic. This was interactive Web application for testing, which used services from Apache WebServer.
The application which I developed were running on Node Server and on the OAM/OSS.
I supported development of test tools/ utilities and scripts using (Perl, Tcl,awk, Unix Shell) on these projects.
Clearcase with Multisite environment were used for SCM. ClearQuest tool was used for the bug report Generations/Tracking of appropriate Action Requests. The quality of the SW was produced according to the CCM, and internal Software Process Improvements(SPI) guidelines. Rational Suite Componets were used in all development and test phases.
The Technical Specifications, for which I was responsible, were the "core" architecture of the Access Node which was developed. One of the concept from the Technical specification was submitted by NOKIA NET for a Patent as part of the "Invention Report".

Year: 10/1990 – 08/1996
Customer: ALCATEL

Project Description: Alcatel SEL - Stuttgart, Germany, October 1990 to March 1993.
I was Software Engineer on SDH, PDH Cross Connect project whose applications were, both Network Management and Telecommunication (4/3/1 cross connects). My role was to provide SW services for Cross-Connection, using the Termination Points. These were represented as Managed Objects, according to the CM-Information Model, as specified by the Deutsche Bundespost Telekom.
For the Network Management, I was involved with: TMN, ROSE, CMISE, ASN.1, GDMO related Applications, Alarm Management, Performance Management, Configuration Management, and Cross Connect Management. The design was produced using the OOD, OOP, with and GNU C++. All the documentation was produced using the CCITT (ITU),IEEE, ETSI, ISO international Standards.

Alcatel BELL, Antwerp, from Jan 94 to Jan 95.
I was Project Manager for the Alcatel Network Integrated Management System project responsible for: full technical responsibility, together with Project Planning, Budget Control (using MS Project). Fixed fee task specification, Estimation, Project Reporting, Tracking , Risk Evaluation, Contingency Planning and Execution.
Alcatel Telecommunication Systems -ATS, Den Haag (Jan 95 to August 96).
Provide Network Management SW suport for Provisioning of Data Collection for a Mediation Device, for Fixed Access and IP Networks.
Designed a Data Persistency Solution for the applications, using the Relation database (Oracle7.1); this was implemented by a generic C++ base class, which is inherited by the applications which need to make their data persistent. I also, examined an Object Oriented data base (OODBMS) as a possible solution, affecting the TMN applications in the near future.
Provided support for Software Process Improvement (SPI), and Software Quality Management, the documentation were in accordance to the ISO9000, IEEE, and CMM (Capability Maturity Model). Furthermore, specified the design and test templates, and the following guidelines: Test procedures, GDMO, Mapping of GDMO to Booch , Software Development Process, and the C++ Guidelines.